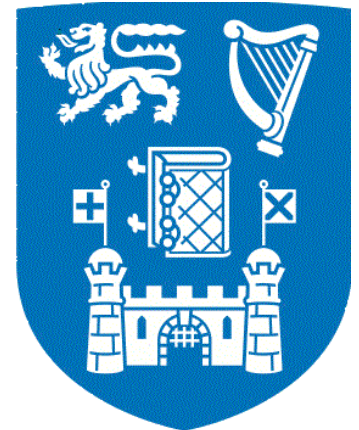


Formal Verification CSU44004/CSU55004

Lecture 4: Propositional Logic

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Quiz

Q1: Show if the following holds or not using indirect proof method.

$$(p \rightarrow (q \vee r)), (q \rightarrow r), p \models (\neg r \rightarrow s)$$

Q2: Using semantic reasoning (truth values), determine whether the following entailment holds: $(p \rightarrow q), (q \rightarrow r), p \models r$.

Bonus Question

Q3: which of the following are properties of $(p \rightarrow \neg q) \rightarrow (q \vee \neg p)$?

- A) Satisfiable
- B) Unsatisfiable
- C) Valid
- D) Invalid

Q1: Sol

Assuming all premises are true:

$$1. p \rightarrow (q \vee r) = T$$

$$2. q \rightarrow r = T$$

$$3. p = T$$

Since p is true (3), $q \vee r$ must be true (from premise 1).

Premise 2 says: $q \rightarrow r$

So, if q is true, r must be true.

Therefore, regardless of whether q is true or false, we get: $r = \text{true}$

Why?

If q is true $\rightarrow r$ is true.

If q is false $\rightarrow$ since $q \vee r$ is true, r must be true.

The conclusion is: $\neg r \rightarrow s$

But we know: $r = \text{true}$

Therefore: $\neg r = \text{false}$

An implication with a false antecedent is true:

$$\text{False} \rightarrow s = \text{True}$$

regardless of the value of s .

So, it holds.

Q2: Sol

p	q	r	$p \rightarrow q$	$q \rightarrow r$	Premises all true?	Conclusion r
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	F	T
T	F	F	F	T	F	F
F	T	T	T	T	F	T
F	T	F	T	F	F	F
F	F	T	T	T	F	T
F	F	F	T	T	F	F

Notice that there is only one row where all premises are true and there is no row where all premises are true and the conclusion is false. So, entailment holds.

p	q	r	$p \rightarrow q$	$q \rightarrow r$
T	T	T	T	T

Recap

- ❖ Propositional logic formulas are syntax:
 - ❖ $A, C, B ::= p \mid (\neg A) \mid (A \wedge A) \mid (A \vee A) \mid (A \rightarrow A)$
- ❖ A ***valuation/model*** of A is an assignment of **T** or **F** to (at least) the atomic propositions in A .
- ❖ Given a model M of A , the semantics of A for this model is either **T** or **F**
 - ❖ “ M makes A **T** (or **F**)”
 - ❖ we construct the truth table of A to see which one it is
 - ❖ A can be *satisfiable/falsifiable/valid/invalid*
- ❖ ***Semantic entailment***: $A_1, \dots, A_n \models B$ is the statement
 - ❖ Any model M making all $A_1, \dots, A_n$ **T**, also makes B **T**.

Semantic entailment

- ❖ To check whether the following is a correct entailment
 - ❖ $(p \rightarrow (q \vee r)), (q \rightarrow r), p \models (\neg r \rightarrow s)$
- ❖ we need to construct and check 16 lines in a truth table ($2^{\text{\#atomic propositions}}$).
- ❖ Brute force algorithm for checking semantic entailment is exponential to the number of atomic propositions.
- ❖ an *indirect proof* can be quicker.

Indirect Proofs of semantic entailment

- ❖ Indirect proof:
 - ❖ 1. assume entailment is *falsifiable/Incorrect*
 - ❖ 2. check if that's possible
- ❖ Falsifiable iff there is a valuation making all premises T and the conclusion F.
- ❖ Example: check if the following entailment holds:
 - ❖ $(p \rightarrow (q \vee r)), (q \rightarrow s), p \models (\neg r \rightarrow s)$
 $T \quad , \quad T \quad , T \models F$
- ❖ **Contradiction:** q has to be both T and F. Therefore, the above is not falsifiable. Therefore, the entailment holds.

Syntactic Logical Proofs

- ❖ Syntactic logical proofs are proofs where we derive conclusions from premises by applying formal inference rules, without using truth tables or checking valuations.
- ❖ Example: If it is raining, then the ground is wet. $(p \text{ implies } q), p$
 - ❖ It is raining. q
 - ❖ Therefore, the ground is wet.
- ❖ Example: Dublin is in Ireland.
 - ❖ Paris is in France. p, q
 $q \text{ and } q$
 - ❖ Therefore, Dublin is in Ireland and Paris is in France.

Syntactic Logical Proofs

- ❖ Three different entailments (implications)

- ❖ $A \rightarrow B$

- ❖ This is merely syntax, although we have assigned a meaning to this syntax via $sem(A \rightarrow B)$.

- ❖ $A1, \dots, An \models B$

- ❖ Any model M making all A_i **T**, also makes B **T**
- ❖ We can show as a lemma that $\models A \rightarrow B$ iff $A \models B$.

- ❖ $A1, \dots, An \vdash B$

- ❖ “from $A1, \dots, An$ we can syntactically prove B ”
- ❖ “ $A1, \dots, An$ proves B ”
- ❖ i.e., there is a way to start from the formulas $A1, \dots, An$ and derive B using the *inference rules of propositional logic*

Natural Deductions

- ❖ Inference rules
- ❖ *Axiom* (structure of an inference rule) 公理 (推理规则的结构)
- ❖ **In Algebra:** If $a=b$ and $b=c$, then conclude $a=c$.
- ❖ **In logic:** If you have $A1$ and $A2$, then conclude $A1 \wedge A2$.

Examples:

- ❖ ***Axiom ($\wedge i$) (Introduction)***

- ❖ If we have any formulas $A1$ and $A2$ then we can derive the formula $A1 \wedge A2$

$$A1, A2 \vdash A1 \wedge A2$$

- ❖ ***Axiom ($\wedge e1$) (Elimination 1)***

- ❖ If we have formula $A1 \wedge A2$ then we can derive the formula $A1$

$$A1 \wedge A2 \vdash A1$$

- ❖ ***Axiom ($\wedge e2$) (Elimination 2)***

- ❖ If we have formula $A1 \wedge A2$ then we can derive the formula $A2$

$$A1 \wedge A2 \vdash A2$$

Natural Deductions

- ❖ There is a standard calculus for axioms of this form called Natural Deduction
- ❖ Structure of inference axioms:

$$\frac{A1, \dots, An}{B} \text{ Rule Name}$$

- ❖ Examples:

$$\frac{A1 \quad A2}{A1 \wedge A2} \wedge i$$

Suppose we know: p and q
Then we may derive: p ∧ q.
Proof:
1. p
2. q
3. p ∧ q (∧ i, 1, 2)

$$\frac{A1 \wedge An}{A1} \wedge e1$$

Given: p ∧ q
Then we can derive: p.
Proof:
1. p ∧ q
2. p (∧ e1)

$$\frac{A1 \wedge An}{A2} \wedge e2$$

Given: p ∧ q
Then we can derive: q.
Proof:
1. p ∧ q
2. q (∧ e2)

Natural Deductions Proofs

自然演绎证明

❖ Prove $(p \wedge q) \wedge r, s \wedge t \vdash q \wedge s$.

❖ Proof:

<u>Line#</u>	<u>Formula</u>	<u>Justification</u>
1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.	$p \wedge q$	$\wedge e_1$ line 1
4.	q	$\wedge e_2$ line 3
5.	s	$\wedge e_1$ line 2
6.	$q \wedge s$	$\wedge i$ line 4,5

Natural Deductions Proofs

- ❖ Modus Ponens ($\rightarrow$ e)

- ❖ From: $A, A \rightarrow B$

- ❖ derive: B

- ❖ A: it rained

- ❖ $A \rightarrow B$: if it rained then the street is wet.

- ❖ we derive: B: the street is wet

- ❖ Example:

- ❖ Show: $p \rightarrow q \vdash q$

- ❖ Proof:

- ❖ 1. $p \rightarrow q$ Premise

- ❖ 2. p Premise

- ❖ 3. q $\rightarrow$ e (1,2)

Natural Deductions Proofs

❖ Modus Ponens ($\rightarrow$ e)

❖ Example 2:

❖ Premises: $p \rightarrow q$

$q \rightarrow r$

p

❖ Goal: Show r .

❖ Example 3:

❖ Premises: $p \wedge q$

$p \rightarrow r$

❖ Goal: Show r .

1	$p \rightarrow q$	Premise
2	$q \rightarrow r$	Premise
3	p	Premise
4	q	$\rightarrow$ e from (1,3)
5	r	$\rightarrow$ e from (2,4)

1	$p \wedge q$	Premise
2	$p \rightarrow r$	Premise
3	p	$\wedge e_1$ from (1)
4	r	$\rightarrow$ e from (2,3)

Natural Deductions Proofs

❖ Modus Tollens (MT)

❖ If A implies B, and B is false, then A must also be false.

❖ A: it rained

❖ B: streets are wet.

❖ Premises: $A \rightarrow B$: if it rained then the street is wet.

❖ The street is not wet. ($\neg B$)

❖ Conclusion: Therefore, it did not rain. ($\neg A$)

$$\frac{A \rightarrow B \quad \neg B}{\neg A} MT$$

Natural Deductions Proofs

❖ Implication Introduction ($\rightarrow I$)

❖ Implication Introduction is the rule that lets you prove a statement of the form $A \rightarrow B$.

❖ The idea is:

❖ Assume A , prove B , then conclude $A \rightarrow B$

❖ Assume its raining

❖ Show that the street is wet.

❖ Conclude: if its rains then street is wet.

❖ Example:

❖ Show: $p \vdash q \rightarrow (p \wedge q)$

❖ Proof:

- | | | | |
|---|----|------------------------------|-----------------------|
| ❖ | 1. | p | Premise |
| ❖ | 2. | q | Assumption |
| ❖ | 3. | $p \wedge q$ | $\wedge I$ (1,2) |
| ❖ | 4. | $q \rightarrow (p \wedge q)$ | $\rightarrow I$ (2,3) |

$$\frac{\begin{array}{c} [A] \\ \vdots \\ B \end{array}}{A \rightarrow B} (\rightarrow I)$$

Natural Deductions Proofs

❖ Disjunction Introduction(Vi)

$$\text{❖ } \frac{A1}{A1 \vee A2} \vee i1$$

$$\text{❖ } \frac{A2}{A1 \vee A2} \vee i2$$

❖ Disjunction Elimination (Ve)

$$\text{❖ } \frac{A1 \vee A2 \quad A1 \vdash A3 \quad A2 \vdash A3}{A3} \vee e$$

❖ Example:

❖ Premises: $p \vee q$

$p \rightarrow r$

$q \rightarrow r$

❖ Show r .

Proof idea

- If p is true, then r is true.
- If q is true, then r is true.
- Since one of p or q must be true, r is true.

1.	$(p \vee q)$	premise
2.	$p \rightarrow r$	premise
3.	$q \rightarrow r$	$\wedge e_1$ line 1
4.	Assume p	
5.	r	$\rightarrow e$ line 2,4
6.	Assume q	
7.	r	$\rightarrow e$ line 3,6
8.	r	$\vee e$ 1,4 – 5,6 – 7